

# ASTR 201 — Midterm 2 Exam Solutions

Astronomy for Science Majors — Spring 2026

San Diego State University

Instructor: Dr. Anna Rosen

---

## Instructions:

- There are **two parts** to this exam, worth **50 points** total. Answer all questions in the space provided.
- Show your work in a clear and orderly fashion, including all unit conversions, for full credit on calculation-based questions. **Circle your final answers clearly.**
- Partial credit will be awarded where appropriate.
- A formula sheet is provided; identify and use the appropriate equations where necessary. CGS units are used throughout.
- Use **proportional reasoning, ratio methods**, and scaling arguments wherever possible — full derivations are not required unless specifically requested.

## Part I: Multiple Choice Questions

Please show your work in the space provided if necessary and provide your answer on the line provided. (14 points, 2 points each)

1. For a main-sequence star, which of the following is the correct ordering of the three fundamental stellar timescales from **shortest to longest**?

- (a)  $\tau_{\text{nuc}} < \tau_{\text{KH}} < \tau_{\text{dyn}}$
- (b)  $\tau_{\text{KH}} < \tau_{\text{dyn}} < \tau_{\text{nuc}}$
- (c)  $\tau_{\text{dyn}} < \tau_{\text{KH}} < \tau_{\text{nuc}}$
- (d)  $\tau_{\text{dyn}} < \tau_{\text{nuc}} < \tau_{\text{KH}}$
- (e) All three are comparable on the main sequence.

Answer: \_\_\_\_\_

**Answer: (c)**

The **dynamical timescale**  $\tau_{\text{dyn}} \sim 1/\sqrt{G\rho} \approx 30 \text{ min}$  (Sun) is set by free-fall under gravity alone — the shortest. The **Kelvin-Helmholtz timescale**  $\tau_{\text{KH}} \sim GM^2/(RL) \approx 30 \text{ Myr}$  (Sun) is the time to radiate away the star's gravitational binding energy. The **nuclear timescale**  $\tau_{\text{nuc}} \sim \epsilon f Mc^2/L \approx 10 \text{ Gyr}$  (Sun) is set by available fusion fuel.

Hierarchy:  $\tau_{\text{dyn}} \ll \tau_{\text{KH}} \ll \tau_{\text{nuc}}$ . The large separation is why stars can be treated as in hydrostatic equilibrium on nuclear-burning timescales.

2. A  $10 M_{\text{sun}}$  main-sequence star has a lifetime approximately how long compared to the Sun's?

- (a)  $10 \times$  longer ( 100 Gyr).
- (b)  $\approx 3 \times$  shorter ( 3 Gyr).
- (c)  $\approx 300 \times$  shorter ( 30 Myr).
- (d)  $\approx 3 \times 10^5 \times$  shorter ( 30 kyr).
- (e) Its lifetime is independent of mass.

Answer: \_\_\_\_\_

**Answer: (c)**

**Main-sequence lifetime scaling:**  $\tau_{\text{MS}} \propto M/L \propto M/M^{3.5} = M^{-2.5}$ .

$$\tau_{\text{MS}} \frac{10M_{\text{sun}}}{\tau_{\text{MS}}} (M_{\text{sun}}) = (10)^{-2.5} \approx \frac{1}{316} \approx 3 \times 10^{-3}$$

With  $\tau_{\text{MS,sun}} \approx 10 \text{ Gyr} \Rightarrow \tau_{\text{MS}}(10M_{\text{sun}}) \approx 30 \text{ Myr}$ . More massive stars burn fuel vastly faster because  $L$  rises faster than  $M$ .

3. If mass is gradually added to a main-sequence star (assume  $R$  stays roughly constant), how does the core respond to maintain hydrostatic equilibrium?

- (a) Core expands;  $T_c$  decreases;  $P_c$  decreases.
- (b) Core contracts;  $T_c$  decreases;  $P_c$  increases.
- (c) Core expands;  $T_c$  increases;  $P_c$  decreases.
- (d) Core contracts;  $T_c$  increases;  $P_c$  increases.
- (e) Core properties remain unchanged.

Answer: \_\_\_\_\_

**Answer: (d)**

More mass at fixed  $R$  increases the inward gravitational force, so the core must support a greater weight. From the scaling relations on the formula sheet:

$$P_c \sim GM^2/R^4 \implies P_c \uparrow \text{ when } M \uparrow$$

$$T_c \sim \mu GMm_p/(k_B R) \implies T_c \uparrow \text{ when } M \uparrow$$

$$\rho_c \sim M/R^3 \implies \rho_c \uparrow \text{ when } M \uparrow$$

To deliver the required  $P_c$ , the core must **contract** ( $\rho_c \uparrow$ ), which heats it ( $T_c \uparrow$ ).

4. Quantum tunneling allows protons to fuse in the Sun's core, even though the classical thermal energy ( $k_B T_c \approx 1$  keV) is far below the Coulomb barrier ( $E_C \approx 1$  MeV). Why is this possible?

- (a) Protons have zero size and so do not feel the Coulomb barrier.
- (b) Gravity is strong enough at the core to push protons together.
- (c) Protons behave as quantum waves, giving a nonzero probability of penetrating the Coulomb barrier.
- (d) Magnetic fields in the core channel protons toward one another.
- (e) The CNO cycle lowers the temperature threshold for fusion.

Answer: \_\_\_\_\_

**Answer: (c)**

At  $T_c \approx 1.5 \times 10^7 \text{ K}$ ,  $k_B T \approx 1$  keV — three orders of magnitude below the Coulomb barrier ( $\sim 1$  MeV). Classically, protons never fuse. But each proton has a de Broglie wavelength  $\lambda_{\text{dB}} = h/p$  comparable to the nuclear-approach distance, so the proton has a nonzero **amplitude** to be found on the other side of the barrier. The Gamow factor  $\Gamma \propto e^{-2G}$  quantifies this tunneling probability.

5. Fusion of elements heavier than iron ( $^{56}\text{Fe}$ ) does not release energy in stellar cores. Why?

- (a) The Coulomb barrier becomes infinitely large for  $Z > 26$ .
- (b) Nuclei heavier than iron have lower binding energy per nucleon, so fusing them absorbs rather than releases energy.
- (c) All available fuel is locked up in iron.
- (d) Neutrino losses prevent further reactions.
- (e) Heavy nuclei spontaneously fission before they can fuse.

Answer: \_\_\_\_\_

**Answer: (b)**

The binding energy per nucleon peaks near  $^{56}\text{Fe}$ . For  $A < 56$ , fusion **releases** energy (exothermic,  $\Delta m > 0$ ). For  $A > 56$ , fusion **absorbs** energy — so it cannot be a stellar energy source. Iron is the “ash” of stellar burning.

6. Light travels from the Sun's core to its surface in roughly 100,000 years, even though light travels at speed  $c$  and  $R_{\text{sun}} \approx 7 \times 10^{10}$  cm (a light-crossing time of  $\sim 2\text{s}$ ). What is the physical reason?

- (a) Photons slow down when passing through dense plasma.
- (b) The photons lose energy and are recreated at each absorption.
- (c) Photons undergo a random walk, being absorbed and re-emitted  $\sim 10^{10}$  times, so energy diffuses outward rather than streaming.
- (d) Gravitational time dilation slows photons near the core.
- (e) Convection carries photons slowly from the core.

Answer: \_\_\_\_\_

**Answer: (c)**

Photon mean free path in the solar core is  $\ell = 1/(\kappa\rho) \approx 0.02$  cm. To traverse  $R_{\text{sun}} = 7 \times 10^{10}$  cm by random walk requires  $N \sim (R/\ell)^2 \approx 10^{24}$  steps, giving a diffusion time  $\sim R^2/(\ell c) \sim 10^5$  yr. Photons travel at  $c$  between scatterings, but the **energy** diffuses slowly because the walk is random.

7. Which is the correct evolutionary sequence for a  $1M_{\text{sun}}$  star **after** it leaves the main sequence?

- (a) Main-sequence turnoff → red giant branch → helium flash → horizontal branch → AGB → planetary nebula → white dwarf.
- (b) Main-sequence turnoff → white dwarf → red giant → planetary nebula.
- (c) Red giant → main sequence → horizontal branch → white dwarf.
- (d) Main-sequence turnoff → AGB → red giant → helium flash → white dwarf.
- (e) Main-sequence turnoff → supernova → neutron star.

**Answer:** \_\_\_\_\_

**Answer: (a)**

Low-mass post-MS evolution: core H is exhausted, the core contracts and H-shell burning begins, the envelope expands up the **red giant branch**. The He core becomes degenerate; when  $T_c$  reaches  $10^8\text{K}$ , **helium flash** ignites triple- $\alpha$  briefly. The star settles on the **horizontal branch** (core He burning). After core He exhausts, H + He shell burning drives the **asymptotic giant branch**; intense mass loss ejects the envelope as a **planetary nebula**, leaving a hot C/O **white dwarf** to cool. Option (e) is for massive stars ( $M \gtrsim 8M_{\text{sun}}$ ), which are not on this exam.

## Part II: Problems

**Tip:** Solve algebraically (using ratio/scaling methods) before plugging in numbers. This minimizes arithmetic errors and earns credit for correct setup even if the final number is off.

**1) Stellar Structure Under a Hypothetical Scenario.** Suppose a main-sequence star doubles its mass while somehow keeping its central temperature  $T_c$  **fixed**. Assume the mean molecular weight  $\mu$  and composition remain constant throughout.

(a) Determine how the star's radius  $R$  must change to keep  $T_c$  fixed. Express  $R_2$  (after) in terms of  $R_1$  (before). **(4 pts)**

**Solution:**

**Given:**  $M_2 = 2M_1$ ;  $\mu$ ,  $m_p$ ,  $G$ ,  $k_B$  unchanged;  $T_{c,1} = T_{c,2}$ .

**Find:**  $R_2/R_1$ .

**Equation:**  $T_c \sim \frac{\mu G M m_p}{k_B R} \implies T_c \propto M/R$

**Steps (ratio method):**

$$\frac{T_{c,2}}{T_{c,1}} = \left(\frac{M_2}{M_1}\right) \left(\frac{R_1}{R_2}\right) = 1$$

Using  $M_2/M_1 = 2$ :

$$2 \cdot \left(\frac{R_1}{R_2}\right) = 1 \implies \frac{R_2}{R_1} = 2$$

**$R_2 = 2 R_1$  — the radius must double.**

**Unit check:**  $T_c$  has units K on both sides; the ratio is dimensionless. ✓

**Sanity check:** Doubling  $M$  alone would **increase**  $T_c$  (hotter core). To cancel that effect and keep  $T_c$  fixed, the star must **expand** proportionally — which it does:  $R$  doubles. ✓

(b) How do the central pressure  $P_c$  and central density  $\rho_c$  change under this scenario? Express  $P_{c,2}/P_{c,1}$  and  $\rho_{c,2}/\rho_{c,1}$  as numerical ratios. **(4 pts)**

**Solution:**

**Equations** (from formula sheet):

$$P_c \sim \frac{GM^2}{R^4} \implies P_c \propto M^2/R^4$$

$$\rho_c \sim \frac{M}{R^3} \implies \rho_c \propto M/R^3$$

**Given:**  $M_2/M_1 = 2$ ,  $R_2/R_1 = 2$  (from part (a)).

**Steps:**

$$\frac{P_{c,2}}{P_{c,1}} = \left(\frac{M_2}{M_1}\right)^2 \left(\frac{R_1}{R_2}\right)^4 = 2^2 \cdot \left(\frac{1}{2}\right)^4 = \frac{4}{16} = \frac{1}{4}$$

$$\frac{\rho_{c,2}}{\rho_{c,1}} = \left(\frac{M_2}{M_1}\right) \left(\frac{R_1}{R_2}\right)^3 = 2 \cdot \left(\frac{1}{2}\right)^3 = \frac{2}{8} = \frac{1}{4}$$

$$P_{c,2} = \frac{1}{4} P_{c,1}, \quad \rho_{c,2} = \frac{1}{4} \rho_{c,1}$$

**Unit check:** Both are ratios of like quantities, so dimensionless. ✓

**Sanity check:** A bigger, fluffier star at the same  $T_c$  has **lower** central pressure and density — consistent with the ideal-gas relation  $P_c \sim \rho_c k_B T_c / (\mu m_p)$ , where  $P_c$  and  $\rho_c$  scale together when  $T_c$  is held fixed. ✓

(c) Is this scenario physically realistic for a main-sequence star? If not, describe what a real star **actually** does when mass is gradually added — how do  $R$ ,  $T_c$ ,  $P_c$ , and the fusion rate respond? Justify your reasoning. **(4 pts)**

**Solution:**

**No, this scenario is not physically realistic** for a main-sequence star.

**Why the scenario fails:** in true hydrostatic equilibrium, doubling  $M$  **requires**  $P_c$  to **increase** as  $P_c \propto M^2/R^4$  to support the added weight. If the star expanded while holding  $T_c$  fixed (as the hypothetical demands), then  $P_c$  would **decrease** (part (b)) — leaving the core unable to support the extra mass against gravity.

**What a real star does:** the core **contracts** (opposite of the hypothetical). From the scalings:

$$M \uparrow, R \downarrow \implies \rho_c \propto M/R^3 \uparrow \quad \text{and} \quad T_c \propto M/R \uparrow$$

So both  $\rho_c$  and  $T_c$  increase. This is a direct consequence of the virial theorem — **gravitational contraction heats the core** (the famous “negative heat capacity” of self-gravitating systems).

**Fusion response:** since the pp-chain rate scales steeply with temperature ( $\varepsilon \propto \rho T^4$ ), even a modest rise in  $T_c$  dramatically accelerates fusion — which raises  $L$  until a new hydrostatic equilibrium is found, with the star settling at a new point higher up the main sequence. This self-regulation is precisely why fusion is stable: any perturbation that heats the core increases the luminosity, which lets the core radiate away the excess energy.



**2) Fusion Sensitivity & The Radiation Limit.** In this problem you will investigate two effects that together shape the upper end of the stellar mass distribution: the temperature sensitivity of hydrogen fusion and the limit set by radiation pressure.

(a) Consider the pp-chain hydrogen-fusion rate per unit mass,  $\varepsilon_{\text{pp}}$ , deep in a stellar core. Suppose the core density **doubles** and the core temperature increases by 50% (i.e.,  $T \rightarrow 1.5T$ ). By what factor does  $\varepsilon_{\text{pp}}$  change? **(4 pts)**

**Solution:**

**Given:**  $\varepsilon_{\text{pp}} \propto \rho T^4$ ;  $\rho \rightarrow 2\rho$ ;  $T \rightarrow 1.5T$ .

**Find:**  $\varepsilon_{\text{new}}/\varepsilon_{\text{old}}$ .

**Steps (ratio method):**

$$\frac{\varepsilon_{\text{new}}}{\varepsilon_{\text{old}}} = \frac{(2\rho)(1.5T)^4}{\rho T^4} = 2 \cdot (1.5)^4 = 2 \cdot 5.0625 \approx 10.1$$

$$\varepsilon_{\text{new}} \approx 10 \varepsilon_{\text{old}} \text{ — the fusion rate increases by a factor of } \sim 10.$$

**Unit check:** All factors are dimensionless ratios. ✓

**Sanity check:** Even a modest 50% temperature rise — with the density only doubling — increases fusion by an order of magnitude. The  $T^4$  dependence is what makes stellar fusion exquisitely temperature-sensitive and self-regulating. ✓

(b) For a  $M = 100M_{\text{sun}}$  main-sequence star with a fully ionized hydrogen envelope, compute the ratio  $L_{\text{MS}}/L_{\text{Edd}}$ . Is this star super-Eddington ( $L_{\text{MS}} > L_{\text{Edd}}$ ) or sub-Eddington ( $L_{\text{MS}} < L_{\text{Edd}}$ )? **(4 pts)**

**Solution:**

**Given:**  $M = 100M_{\text{sun}}$ ; fully ionized H envelope (so  $\kappa$  is the same as the Sun's —  $\kappa_{\text{es}}$  — and cancels). From the formula sheet:  $L_{\text{MS}} \propto M^{3.5}$ ,  $L_{\text{Edd}} \propto M/\kappa$ ,  $L_{\text{Edd,sun}} \approx 3.9 \times 10^4 L_{\text{sun}}$ .

**Find:**  $L_{\text{MS}}/L_{\text{Edd}}$  for this star.

**Steps (ratio method):** Combined scaling (at fixed  $\kappa$ ):

$$\frac{L_{\text{MS}}}{L_{\text{Edd}}} \propto \frac{M^{3.5}}{M} = M^{2.5}$$

Using the Sun as reference:

$$\frac{(L_{\text{MS}}/L_{\text{Edd}})_{\text{star}}}{(L_{\text{MS}}/L_{\text{Edd}})_{\text{sun}}} = \left( \frac{M}{M_{\text{sun}}} \right)^{2.5} = (100)^{2.5} = 10^5$$

For the Sun:

$$(L_{\text{MS}}/L_{\text{Edd}})_{\text{sun}} = L_{\text{sun}}/L_{\text{Edd,sun}} = 1/(3.9 \times 10^4) \approx 2.6 \times 10^{-5}$$

Combining:

$$\left( \frac{L_{\text{MS}}}{L_{\text{Edd}}} \right)_{\text{star}} \approx 10^5 \times (2.6 \times 10^{-5}) \approx 2.6$$

$L_{\text{MS}}/L_{\text{Edd}} \approx 2.6 > 1$  — the star is **super-Eddington**.

**Unit check:** Ratio of luminosities is dimensionless. ✓

**Sanity check:**  $L_{\text{MS}}/L_{\text{Edd}} \propto M^{2.5}$  grows rapidly with mass; the Sun is **deeply** sub-Eddington ( $L_{\text{sun}}/L_{\text{Edd,sun}} \sim 10^{-5}$ ), but by  $100M_{\text{sun}}$  the ratio crosses unity — consistent with the observed upper stellar mass limit near  $\sim 100\text{--}150M_{\text{sun}}$ . ✓

(c) Explain **why very massive stars ( $M \gtrsim 100M_{\text{sun}}$ ) are rare and short-lived**. Use your results from parts (a) and (b) to justify both claims quantitatively. **(4 pts)**

**Solution:**

**Massive stars are short-lived** because fusion rates scale steeply with core temperature (pp-chain  $\varepsilon \propto T^4$ ; CNO  $\varepsilon \propto T^{17}$ ). Higher-mass stars have hotter cores, so they burn through their hydrogen fuel enormously faster than the Sun —  $\tau_{\text{MS}} \propto M^{-2.5}$  implies a  $100M_{\text{sun}}$  star lives only  $\sim 10 \text{ Gyr} \times (100)^{-2.5} \approx 100 \text{ kyr}$ .

**Massive stars are rare** because they are **super-Eddington** (part (b)): radiation pressure on electron-scattering opacity exceeds gravity, driving violent line-driven winds that shed the envelope on timescales comparable to the nuclear lifetime. Above  $\sim 100\text{--}150M_{\text{sun}}$ , the star cannot hold itself together long enough to form in the first place — this sets the observed upper mass cutoff.

Together, the  $T$ -sensitivity of fusion and the Eddington limit conspire to make high-mass stars both cosmically rare **and** fleeting.

**3) The Chandrasekhar Mass & White Dwarf Structure.** When gravity fully compresses the electrons in a white dwarf until they become relativistic, a fundamental mass scale emerges that can be built from only three fundamental constants —  $\hbar$ ,  $c$ ,  $G$  — together with the mass of a proton  $m_p$ .

(a) Using **dimensional analysis**, construct a combination of  $\hbar$ ,  $c$ ,  $G$ , and  $m_p$  that has the units of mass and is expected to yield the Chandrasekhar mass scale. Evaluate your expression numerically (in grams and in  $M_{\text{sun}}$ ). Show that it yields a value of order  $\sim 1M_{\text{sun}}$ . **(4 pts)**

**Solution:**

**Given:**  $\hbar = 1.05 \times 10^{-27} \text{ erg s}$ ;  $c = 3.0 \times 10^{10} \text{ cm s}^{-1}$ ;  $G = 6.67 \times 10^{-8} \text{ cm}^3 \text{ g}^{-1} \text{ s}^{-2}$ ;  $m_p = 1.67 \times 10^{-24} \text{ g}$ ;  $M_{\text{sun}} = 2.0 \times 10^{33} \text{ g}$ .

**Find:** A combination of these constants with units of mass [g].

**Dimensional analysis:** In base CGS:

$$[\hbar c] = \text{erg cm} = \text{g cm}^3 \text{s}^{-2}$$

$$[G] = \text{cm}^3 \text{g}^{-1} \text{s}^{-2}$$

$$[\hbar c/G] = \text{g cm}^3 \frac{\text{s}^{-2}}{\text{cm}^3 \text{g}^{-1} \text{s}^{-2}} = \text{g}^2$$

So  $\sqrt{\hbar c/G}$  has units of mass — this is the **Planck mass**,  $M_{\text{Pl}}$ . To build a larger mass scale involving  $m_p$ , use the combination  $M_{\text{Ch}} \sim M_{\text{Pl}}^3/m_p^2$ :

$$M_{\text{Ch}} \sim \frac{(\hbar c/G)^{3/2}}{m_p^2} [\text{g}]$$

**Numerical evaluation:**

$$\hbar c/G = \frac{3.15 \times 10^{-17} \text{ erg cm}}{6.67 \times 10^{-8} \text{ cm}^3 \text{g}^{-1} \text{s}^{-2}} = 4.72 \times 10^{-10} \text{ g}^2$$

$$M_{\text{Pl}} = \sqrt{4.72 \times 10^{-10}} \text{ g} = 2.17 \times 10^{-5} \text{ g}$$

$$M_{\text{Ch}} \sim \frac{(2.17 \times 10^{-5})^3}{(1.67 \times 10^{-24})^2} \text{ g}$$

$$= \frac{1.02 \times 10^{-14}}{2.79 \times 10^{-48}} \text{ g} \approx 3.7 \times 10^{33} \text{ g}$$

In solar units:

$$\frac{M_{\text{Ch}}}{M_{\text{sun}}} \approx \frac{3.7 \times 10^{33}}{2.0 \times 10^{33}} \approx 1.8$$

$$M_{\text{Ch}} \sim 3.7 \times 10^{33} \text{ g} \approx 1.8 M_{\text{sun}}$$

**Unit check:**  $[M_{\text{Ch}}] = \text{g}^3/\text{g}^2 = \text{g}$ . ✓

**Sanity check:** The accepted value is  $M_{\text{Ch}} \approx 1.4 M_{\text{sun}}$ . Dimensional analysis is expected to match the order of magnitude but miss the  $\mathcal{O}(1)$  prefactor (here, a factor involving the electron fraction  $Y_e$  and numerical coefficients). Agreement within  $\sim 30\%$  is excellent. ✓

(b) Two (non-relativistic) white dwarfs have masses  $M_1 = 0.6M_{\text{sun}}$  and  $M_2 = 1.2M_{\text{sun}}$ . Determine (i) which has the larger **radius** and by what factor, and (ii) which has the larger **average density** and by what factor. **(4 pts)**

**Solution:**

**Given:**  $M_1 = 0.6M_{\text{sun}}$ ,  $M_2 = 1.2M_{\text{sun}}$ . From the formula sheet:  $R_{\text{WD}} \propto M^{-1/3}$  (non-relativistic degenerate matter); average density  $\rho \propto M/R^3$ .

**Find:**  $R_1/R_2$  and  $\rho_1/\rho_2$ .

**Steps — radius ratio (ratio method):**

$$\frac{R_1}{R_2} = \left( \frac{M_1}{M_2} \right)^{-1/3} = \left( \frac{M_2}{M_1} \right)^{1/3} = (2)^{1/3} \approx 1.26$$

$R_1 \approx 1.26 R_2$  — the **less-massive WD ( $M_1$ )** has the **larger radius**.

**Steps — density ratio:** Combine the scalings:

$$\rho \propto \frac{M}{R^3} \propto \frac{M}{(M^{-1/3})^3} = \frac{M}{M^{-1}} = M^2$$

So  $\rho_{\text{WD}} \propto M^2$  (more massive WDs are denser and also smaller). Then

$$\frac{\rho_1}{\rho_2} = \left( \frac{M_1}{M_2} \right)^2 = \left( \frac{0.6}{1.2} \right)^2 = \left( \frac{1}{2} \right)^2 = \frac{1}{4}$$

$\rho_1 = \frac{1}{4} \rho_2$  — the **more-massive WD ( $M_2$ )** is **4 × denser**.

**Unit check:** Both are dimensionless ratios of like quantities. ✓

**Sanity check:** The WD mass-radius relation is **inverted** — adding mass makes the star smaller. Since density piles up both from more mass **and** less volume,  $\rho \propto M^2$  grows quickly. As  $M \rightarrow M_{\text{Ch}} \approx 1.4M_{\text{sun}}$ ,  $\rho \rightarrow \infty$  in the non-relativistic limit — which is precisely where that limit breaks down (relativistic electrons, part (c)). ✓

(c) The Chandrasekhar mass  $M_{\text{Ch}} \approx 1.4M_{\text{sun}}$  is an **upper limit** on white-dwarf masses: no stable WD can exceed it. Explain **why this limit exists** physically, and justify your reasoning quantitatively. **(4 pts)**

**Solution:**

As a WD is compressed, electrons are packed more tightly; by the uncertainty principle their momenta — and hence their speeds — grow. Below  $\sim 1.4M_{\text{sun}}$ , electron speeds remain non-relativistic ( $v \ll c$ ) and degeneracy pressure scales steeply as  $P_{\text{deg}} \propto \rho^{5/3}$ , rising fast enough to resist gravity.

When the density is so high that electrons become **relativistic** ( $v \rightarrow c$ ), their speeds approach the cosmic speed limit even though their momenta can still increase. The pressure law therefore softens to  $P_{\text{deg}} \propto \rho^{4/3}$ . Crucially, **the rate at which  $P_{\text{deg}}$  grows with added mass can no longer outpace the rate at which gravity grows**: doubling the mass increases gravitational compression faster than the relativistic degeneracy pressure can respond. No stable equilibrium exists — the star collapses to a neutron star or black hole.

Physically,  $M_{\text{Ch}}$  marks the mass at which electrons are forced to become fully relativistic before they can support the star.

— End of Exam —

Formula sheet provided separately.